INFAN Image Deblur - Some Notes for Photographers

The top left below is a cutout portrait of my favourite chicken/escape artist. It was taken with my 1957 Jupiter 9 85mm lens. Because she doesn't listen to me when I order her 'Stay Still' there is some motion and out of focus blur. No matter what I did and what type of sharpening I did in affinity it just didn't look natural.
I took the cutout of the head and ran it through the deblur program with the result (bottom left)and then used Affinity to merged/blended the enhanced image into the original image with the result on the left side. See the second image, it may not look like much but sometimes I need to elevate an important image without introducing fake AI artifacts.



Top left: original, Bottom left: deblurred, Right side: result

Deblurring vs. Sharpening vs. AI Enhancement

Imagine a satellite trying to read a license plate on the ground. The satellite's motion during the shot and the wavy atmosphere smear each letter into a fuzzy streak. Since you don't know the exact smear, the software estimates the most likely kind of blur that could have caused that fuzziness, reverses it, then adjusts its guess and tries again until the letters come back into focus. The information is still there—just smeared—and deblurring tries to un-smear it using its best guess.
This is different from sharpening, which just makes the fuzzy streaks look punchier, not readable. And it's different from AI enhancing, which might guess a plausible license plate number based on patterns it has seen—but could just as easily guess wrong. Deblurring tries to recover what was actually there; the other two don't.

The image below was a deblur of the full sized image then the result was subsequently merged with the original image to enhance the signs, the people and activity.



The Grand Bazaar with a history of over 5 centuries, Iran (1977 the year the Eaton Centre was opened) Kodak Kodachrome 64

Sometimes as in this case due to a toy film camera your holiday shot turns out to be useless. Using deblur on a social media sized crop can just maybe look good enough to post.



Also sometimes it can be used on a large sized image in place of sharpening. The result is more organic than other methods. But this only works on some images. Apply IFAN to the full size version of the image above results in too many artifacts. Trying to deblur structures at full resolution on an image like above is a pointless task.